S : Introduction to Multiplication

For the tutor: This worksheet introduces multiplication conceptually. Begin with equal grouping using physical objects (stones, bottle caps, beans). If the student struggles, stop and model with objects or drawings.

Section A: Equal Groups Concrete Understanding)

For the tutor: Multiplication means having the same number in each group. Read each prompt aloud. Use drawings or objects if needed.

1. There are **3 groups** with **stones in each group**. How many stones are there altogether?

2. There are **4 groups** with **5 pencils in each group**. How many pencils are there altogether?

3. Draw **5 groups** with **3 circles in each group**. How many circles are there in total?

4. Circle the picture that shows **equal groups**. (Explain your choice.)

Section B: Writing Equal Groups as Multiplication

For the tutor: Help the student say the multiplication aloud (e.g., "3 groups of 4"). Do not introduce times tables yet.

1. 3 groups of 4 = _____
2. 5 groups of 2 = _____
3. 2 groups of 6 = _____
4. 4 groups of 3 = _____

5. Draw a picture for: 3 5

6. Write a multiplication sentence for your drawing.

Section C: Rectangles and Arrays

For the tutor: A rectangle shows rows and columns. Each row must have the same number of objects.

1. Draw a rectangle with **3 rows** and **4 boxes in each row**. How many boxes are there altogether?

2. Draw a rectangle with **rows** and **6 boxes in each row**. Write the multiplication sentence.

3. How is a rectangle the same as equal groups? Explain using words or drawings.

Section D: Count by Groups Skip Counting)

For the tutor: Have the student count aloud by groups. This builds meaning before memorization.

1. Count by 3s:
   3, 6, _____, _____

2. Count by 4s:
   4, 8, _____, _____

3. 4 3 = _____

4. 5 2 = _____

5. Which is easier for you: counting by ones or by groups? Why?

Section E: Word Problems with Equal Groups

Read each problem and decide if multiplication is needed

1. There are 4 bags. Each bag has 3 oranges. How many oranges are there in total? Multiply / Add: _____

2. A farmer plants 5 rows of maize. There are 6 plants in each row. How many plants are there altogether? Multiply / Add: _____

3. There are 10 children. Each child has 2 shoes. How many shoes are there in total? Multiply / Add: _____

Section F: Order in Multiplication

For the tutor: Multiplication order does not change the total, but the picture may look di erent.

1. Draw 2 4 as a rectangle.

2. Draw 4 2 as a rectangle.

3. Are the totals the same? Yes No

4. Explain how the pictures are di erent but the total is the same.